Ultrasonic + Two-Button LED Controller
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What this is

This Arduino sketch is just like the basic two-button LED pattern controller, but now it ALSO listens to an ultrasonic distance sensor (like HC-SR04).
So:
- You can still press Button A to go to the NEXT LED pattern.
- You can still press Button B to go to the PREVIOUS LED pattern.
- AND the LEDs can react when something is close to the sensor (a bottle, a hand, a cart, etc.).

Typical use case

Your lab setup flings or moves objects past a fixed point. When the ultrasonic sensor "sees" (detects) something closer than a certain distance, the sketch turns on / changes an LED, or switches to a "detected" pattern.

High-level flow
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1. Read buttons (to change patterns manually).
2. Read the ultrasonic sensor (to see if something is in front of it).
3. If something is detected (distance < TRIGGER_DISTANCE_CM), run the "detection" behavior (for example, turn LED strip ON or run a special pattern).
4. Otherwise, just run the currently selected pattern (like normal).
5. Repeat.

What you are allowed to change

Only edit the CONFIGURATION SECTION at the top of the .ino file.

You can change:
- LED_PIN : Arduino pin for LED(s) or LED strip
- NUM_LEDS : how many LEDs
- BUTTON_NEXT_PIN : pin for "next pattern"
- BUTTON_PREV_PIN : pin for "previous pattern"
- ULTRASONIC_TRIG_PIN : TRIG pin for the HC-SR04
- ULTRASONIC_ECHO_PIN : ECHO pin for the HC-SR04
- TRIGGER_DISTANCE_CM : how close an object must be to "count" as detected
- PATTERN_DELAY_MS : how fast the normal animations run
- DETECTED_HOLD_MS : how long to hold the detected pattern/LED on

Do NOT change the logic in loop(), the function that reads the ultrasonic sensor, or the button helper functions, unless you know what you are doing.

How ultrasonic works (simple version)

- Arduino sends a short pulse on the TRIG pin.
- The ultrasonic sensor sends out a sound pulse.
- When the sound comes back, the sensor raises the ECHO pin.
- Arduino measures how long the ECHO pin was HIGH.
- The time --> distance using a simple formula.
- If the distance is small (object is close), we say "detected".

Wiring notes

- HC-SR04 has 4 pins: VCC (5V), GND, TRIG, ECHO.
- TRIG goes to ULTRASONIC_TRIG_PIN.
- ECHO goes to ULTRASONIC_ECHO_PIN (use a 10k/4.7k divider if your board
  is 3.3V).
- Buttons are wired to GND and use INPUT_PULLUP in code.
  That means: pressed = LOW.

Customization examples

- "Sensor is too sensitive."   --> decrease TRIGGER_DISTANCE_CM
- "Sensor doesn't trigger soon." --> increase TRIGGER_DISTANCE_CM
- "Detection LED too short."   --> increase DETECTED_HOLD_MS
- "Animation too fast."   --> increase PATTERN_DELAY_MS
- "Wrong pins."   --> change the *_PIN values

Troubleshooting

- Always prints 0 cm or very large number:
  - Check TRIG/ECHO wiring
  - Make sure ECHO is on a pin that can read digital pulses
- Never detects anything:
  - Object might be outside the cone of the sensor
  - TRIGGER_DISTANCE_CM might be too small
- Buttons stopped working:
  - Make sure you did not remove the button code from loop()
  - Make sure buttons are really to GND (INPUT_PULLUP logic)

License / usage

Share it with your classmates. Keep the comments so people know what to change.